

Monty Hall Problem Using Conditional Probability

Assume you initially choose **Door 1**.

Define:

- C_i : the event that the car is behind Door i
- M_3 : the event that Monty opens Door 3 and reveals a goat

After Monty opens Door 3, switching means choosing Door 2. Therefore, we want to compute

$$P(C_2 \mid M_3).$$

Step 1: Prior Probabilities

Since the car is equally likely to be behind any door,

$$P(C_1) = P(C_2) = P(C_3) = \frac{1}{3}.$$

Step 2: Monty's Behavior

Assume Monty:

1. Knows where the car is.
2. Always opens a door containing a goat.
3. If he has two goat doors available, chooses randomly.

Then:

Case 1: Car behind Door 1

You have initially chosen the car.

Both Door 2 and Door 3 contain goats, so Monty chooses randomly between them.

$$P(M_3 \mid C_1) = \frac{1}{2}.$$

Case 2: Car behind Door 2

Door 3 is the only door Monty can open.

$$P(M_3 \mid C_2) = 1.$$

Case 3: Car behind Door 3

Monty cannot open Door 3 because it contains the car.

$$P(M_3 \mid C_3) = 0.$$

Step 3: Compute $P(M_3)$

Using the Law of Total Probability,

$$P(M_3) = P(M_3 \mid C_1)P(C_1) + P(M_3 \mid C_2)P(C_2) + P(M_3 \mid C_3)P(C_3).$$

Substituting the values,

$$P(M_3) = \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}.$$

Step 4: Apply Bayes' Theorem

$$P(C_2 \mid M_3) = \frac{P(M_3 \mid C_2)P(C_2)}{P(M_3)}.$$

Substituting the values,

$$P(C_2 \mid M_3) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

Therefore,

$$\boxed{P(\text{win by switching}) = \frac{2}{3}}.$$

Similarly,

$$P(C_1 \mid M_3) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}.$$

Hence, after Monty opens Door 3:

$$P(\text{car behind your original door}) = \frac{1}{3},$$

$$P(\text{car behind the other unopened door}) = \frac{2}{3}.$$

So switching increases your probability of winning from $1/3$ to $2/3$.